Printed Pages – 3 Roll No. : .....

5100114(24)

M. Tech. (First Semester) Examination Nov.-Dec. 2019

(Old Scheme)

(Elect. Engg. Br.)

(Specialization : High Voltage Engineering)

HIGH VOLTAGE AC/DC TRANSMISSION

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 40

Note : *Attempt all questions. Attempt any* ***two*** *parts from (a), (b) and (c) of each unit. All questions carry equal marks.*

Unit - I

1. (a) Explain about sequential impedance of AC system.

(b) Explain about statistical and conventional method of insulation design for switching over voltages.

(c) Briefly describe about shunt capacitor compensation on receiving end of a long transmission line.

Unit - II

2. (a) What are the different types of DC links? Explain in detail.

(b) Explain regarding the different factors involved in planning for HUDC transmission.

(c) With a neat diagram explain the switching characteristics of Thyristor.

Unit - III

3. (a) Explain about Gnaetz circuit during without overlap condition. https://www.csvtuonline.com

(b) Describe about the characteristics of a twelve pulse converter.

(c) Briefly describe about system control Hierarchy.

Unit - IV

4. (a) Describe about protection of converter against over currents.

(b) Describe about the operation and characteristics of surge arresters.

(c) Explain the transient over voltages in DC line.

Unit - V

5. (a) Describe about the sources of reactive power.

(b) With a neat diagram, explain the operation of Thyristor Switched Capacitor (TSC).

(c) Briefly explain the operation of DC filters.